

CSIE 2344, Spring 2026: Laboratory 2

Due May 25 (Monday) at Noon

There are 48 points in total. When you submit your homework on Gradescope, please select the corresponding page(s) of each question.

1 Part 1 (32pts)

Replace `<index>` in `lab2.v` with proper indexes, but leave the rest of the code unchanged.

1. (12pts) Print out the module `mult_fast`.
2. (12pts) Show the waveform with the settings in `lab2.sav` (should include 1750 to 1800 sec).
3. (8pts) What is the latency (worst case waiting time from the input becomes steady to the register of the last stage refreshes)?

2 Part 2 (16pts)

Minimize the clock cycle by changing the delays in the module `mult_tb`.

1. (8pts) What is the minimum clock cycle?
2. (8pts) Show the waveform with the settings in `lab2.sav` (should include 1750 to 1800 sec).